12/03/2021

Code-A

Aakash
Medical | IIT-JEE | Foundations

Regd. Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

MM : 720

NEET BOOSTER TEST SERIES (NBTS) for NEET-2021

Time : 3.00 Hrs.

Test - I3

Answer Key

1.	(3)	37.	(4)	73.	(1)	109.	(2)	145.	(1)
2.	(2)	38.	(1)	74.	(1)	110.	(2)	146	(3)
3.	(2)	39.	(3)	75.	(3)	111.	(3)	147.	(2)
4.	(4)	40.	(4)	76.	(4)	112.	(3)	148.	(2)
5.	(3)	41.	(4)	77.	(3)	113.	(4)	149.	(2)
6.	(1)	42.	(4)	78.	(2)	114.	(1)	150.	(2)
7.	(3)	43.	(3)	79.	(2)	115.	(4)	151.	(2)
8.	(3)	44.	(1)	80.	(3)	116.	(1)	152.	(4)
9.	(3)	45.	(1)	81.	(1)	117.	(4)	153.	(2)
10.	(2)	46.	(3)	82.	(2)	118.	(3)	154.	(3)
11.	(1)	47.	(3)	83.	(4)	119.	(2)	155.	(1)
12.	(2)	48.	(3)	84.	(2)	120.	(1)	156.	(2)
13.	(1)	49.	(1)	85.	(3)	121.	(4)	157.	(1)
14.	(4)	50.	(2)	86.	(4)	122.	(3)	158.	(2)
15.	(1)	51.	(3)	87.	(3)	123.	(1)	159.	(3)
16.	(2)	52.	(2)	88.	(2)	124.	(2)	160.	(2)
17.	(2)	53.	(2)	89.	(4)	125.	(3)	161.	(3)
18.	(4)	54.	(4)	90.	(2)	126.	(4)	162.	(3)
19.	(2)	55.	(2)	91.	(2)	127.	(2)	163.	(4)
20.	(1)	56.	(4)	92.	(2)	128.	(1)	164.	(1)
21.	(2)	57.	(3)	93.	(2)	129.	(3)	165.	(2)
22.	(3)	58.	(2)	94.	(1)	130.	(3)	166.	(2)
23.	(2)	59.	(1)	95.	(1)	131.	(2)	167.	(3)
24.	(4)	60.	(2)	96.	(4)	132.	(3)	168.	(2)
25.	(1)	61.	(4)	97.	(3)	133.	(4)	169.	(4)
26.	(2)	62.	(1)	98.	(1)	134.	(2)	170.	(3)
27.	(3)	63.	(1)	99.	(3)	135.	(3)	171.	(3)
28.	(1)	64.	(2)	100.	(4)	136.	(2)	172.	(1)
29.	(4)	65.	(2)	101.	(1)	137.	(2)	173.	(3)
30.	(2)	66.	(3)	102.	(4)	138.	(2)	174.	(2)
31.	(4)	67.	(1)	103.	(1)	139.	(3)	175.	(2)
32.	(2)	68.	(2)	104.	(2)	140.	(4)	176.	(2)
33.	(2)	69.	(3)	105.	(3)	141.	(2)	177.	(2)
34.	(4)	70.	(3)	106.	(4)	142.	(3)	178.	(3)
35.	(3)	71.	(3)	107.	(2)	143.	(1)	179.	(3)
36.	(2)	72.	(2)	108.	(3)	144.	(2)	180.	(1)

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MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** Time : 3:00 Hrs.

Test - 13

Hints and Solutions

PHYSICS

1. Answer (3)

$V_x = 0.4\ \text{ms}^{-1}$, $V_y = 0 + 0.15 \times 2 = 0.3\ \text{ms}^{-1}$

$V = \sqrt{V_x^2 + V_y^2} = \sqrt{(0.4)^2 + (0.3)^2} = 0.5\ \text{ms}^{-1}$

2. Answer (2)

Time of ascending = $\frac{8}{2}$ = 4 sec.

In next second it falls by

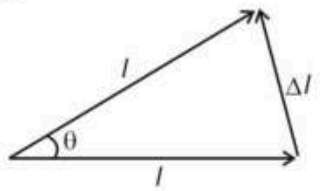
$\Rightarrow h = \frac{1}{2} \times 10 \times (1)^2 = 5\ \text{m}$

Maximum height reached

$= \frac{u^2}{2g} = \frac{40 \times 40}{2 \times 10} = 80\ \text{m}$

Location from ground = 80 – 5 = 75 m.

3. Answer (2)



$\Delta l = \sqrt{l^2 + l^2 - 2l\,l.\cos\theta}$

$= \sqrt{2l^2(1-\cos\theta)}$

$= \sqrt{2l^2 + 2\sin^2\frac{\theta}{2}} = 2l\sin\frac{\theta}{2}$

4. Answer (4)

$x = 5 + 2t^2$

$v = \frac{dx}{dt} = 4t$

$t = 0,\ v = 0$

$t = 2,\ v = 4 \times 2 = 8\ \text{ms}^{-1}$

$\Delta k = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2} \times 2\left[8^2 - 0^2\right] = 64\ \text{J}$

5. Answer (3)

From ground $v = u - gt$

$v = 98 - 9.8 \times 2 = 98 - 19.6 = 78.4\ \text{ms}^{-1}$

Velocity of falling body= $v' = u + gt = 0 + 9.8 \times 2 = 19.6\ \text{ms}^{-1}$

$|p_1| = |p_2|$

$10 \times 78.4 = m \times 19.6$

$\therefore\ m = 40\ \text{kg}$

6. Answer (1)

$F_x = \frac{dp_x}{dt} = \frac{d}{dt}(2t^2 + 6) = 4t$

$F_y = \frac{dp_y}{dt} = \frac{d}{dt}\left(\frac{3t^2}{2} + 3\right) = 3t$

at $t = 2$, $F_x = 4 \times 2 = 8$

$F_y = 3 \times 2 = 6$

$|F| = \sqrt{F_x^2 + F_y^2} = \sqrt{(8)^2 + (6)^2} = 10$ units

7. Answer (3)

Momentum conservation for system

$5 \times 14 + (140 + 60)v = 0$

$$v = -\frac{5\times14}{200} = \frac{-70}{200} = -0.35 \text{ ms}^{-1}$$

8. Answer (3)

$Y = \overline{\overline{A+B}} = A + B \Rightarrow$ OR gate

9. Answer (3)

Solar cell is operated in unbiased condition and its I-V characteristics lies in fourth quadrant. Zener diode acts as voltage regulator when it is reverse biased.

For red LED, the $GaAs_{0.6}P_{0.4}$ material is used for diode fabrication.

10. Answer (2)

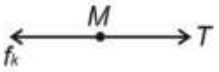
Change in momentum

$= 0.3 \times (12) - (-0.3 \times 12)$

$= 3.6 + 3.6 = 7.2$ N-s

11. Answer (1)

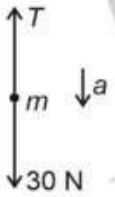
For block F.B.D.



$T - f_k = 20 \times a$

$T - 0.04 \times 200 = 20a$

For hanging block



$30 - T = 3a$

$T - 8 = 20a$

Solving

$$\therefore \quad T = 27.1 \text{ N and } a = \frac{22}{23} \text{ m s}^{-2}$$

12. Answer (2)

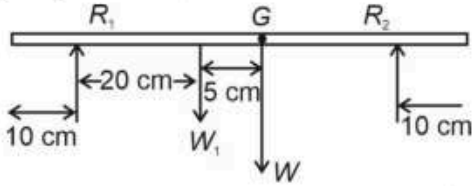
$v = \sqrt{rg\tan\theta} = \sqrt{300\times10\times\tan(15^\circ)}$

$= \sqrt{780} = 27.9 \text{ ms}^{-1}$

13. Answer (1)

For vertical equilibrium

$R_1 + R_2 - W - W_1 = 0$



For rotational equilibrium torque about G $\sum\tau = 0$

$-R_1 \times 0.25 + 0.05 \times W_1 + R_2 \times 0.25 = 0$

$\therefore \quad R_1 + R_2 = 4 \text{ g} + 6 \text{ g} = 10 \text{ g} = 98$ N

$R_1 - R_2 = 1.2 \times 9.8 = 11.76$ N

$\therefore \quad R_1 = 54.88$ N, $R_2 = 43.12$ N

14. Answer (4)

$$\theta = \frac{(\omega_1 + \omega_2)}{2} \times t \quad \text{const. acceleration}$$

$$= \frac{2\pi(52+20)}{2} \times 16 = 2\pi \times 576$$

$$\text{No. of revolutions} = \frac{2\pi\times576}{2\pi} = 576$$

15. Answer (1)

Time to roll down from state of rest

= Time to roll up to plane

$$t = \frac{v_0}{a} = \frac{5\left(1+\frac{1}{2}\right)}{g\sin30^\circ} = \frac{5\times\frac{3}{2}}{10\times\frac{1}{2}} = \frac{3}{2}\text{s}$$

$$\left[\because \ a = \frac{g\sin\theta}{1+\frac{I}{mR^2}}\right]$$

$$T = 2t = \frac{3}{2}\times2 = 3\text{s}$$

16. Answer (2)

By homogeneity

$At = L^0T^0$

$$\therefore \ A = \frac{1}{[t]} = M^0L^0T^{-1}$$

Also $\frac{V_0}{A} = x$

$\therefore V_0 = [x][A]$

$= [L][M^0L^0T^{-1}] = M^0LT^{-1}$

17. Answer (2)

Smaller will be damping force, more will be amplitude of oscillation. Hence answer is (2)

18. Answer (4)

Everybody at all time, at all temperatures emit radiation [except T= 0 K]. This radiation from human body is thermal radiations in infrared region.

19. Answer (2)

$\frac{C_p}{C_v} = 1.4 \qquad \Rightarrow C_p = 1.4\ C_v$

$C_p - C_v = 4150$

$1.4\ C_v - C_v = 4150$

$\therefore C_v = \frac{4150}{0.4} = 10375\,\text{J kg}^{-1}\text{K}^{-1}$

20. Answer (1)

Gravitational potential energy at centre

$= -\frac{3}{2}\frac{GMM}{R}$

Gravitational potential energy at height

$= \frac{-GMM}{R+h} = \frac{-GMM}{2R}$ [$\because\ h = R$]

$\Delta U = \Delta W = \frac{-GMM}{2R} - \left(-\frac{3}{2}\frac{GMM}{R}\right)$

$= \frac{3}{2}\frac{GMM}{R} - \frac{GMM}{2R} = \frac{GMM}{R} = \frac{GM^2}{R}$

21. Answer (2)

$\vec{r} = \vec{r}_2 - \vec{r}_1 = (2\hat{i} + 3\hat{j} + 3\hat{k}) - (\hat{i} + \hat{j} + \hat{k})$

$= \hat{i} + 2\hat{j} + 2\hat{k}$

$|r| = \sqrt{1^2 + 2^2 + 2^2} = 3$

By Coulomb's law

$F = \frac{9\times10^9 \times 3\times10^{-6} \times 3\times10^{-6}}{(3)^2}$

$F = 9 \times 10^{-3}\,\text{N}$

22. Answer (3)

For monoatomic gas $\gamma = \frac{5}{3}$

$TV^{\gamma-1} = \text{const.}$

$T_2 = T_1 \times \left(\frac{V_1}{V_2}\right)^{\gamma-1} = T \times \left(\frac{5.6}{0.7}\right)^{\frac{5}{3}-1} = T \times (8)^{\frac{2}{3}} = 4T$

No. of moles of helium $= \frac{5.6}{22.4} = \frac{1}{4}$

$\Delta W = \frac{nR(T_1 - T_2)}{\gamma - 1} = \frac{\frac{1}{4}R \times (T - 4T)}{\left(\frac{5}{3} - 1\right)} = -\frac{9}{8}RT$

23. Answer (2)

By KVL

$60 = \frac{Q}{2C} + \frac{Q}{C} + \frac{Q}{2C} = \frac{2Q}{C} \quad \therefore Q = 30C$

$V_{xy} = \frac{Q}{C} = \frac{30C}{C} = 30\,\text{V}$

24. Answer (4)

$\frac{B}{B_0} = \frac{r^3}{\left(r^2 + x^2\right)^{3/2}} = \left(\frac{12}{13}\right)^3$

$B = 0.5 \times 10^{-3} \times 0.7865 = 0.393 \times 10^{-3}\,\text{T}$

25. Answer (1)

Resistance of upper arm = 5 + 2 = 7 Ω

Resistance of lower arm = 15 + 6 = 21 Ω

$\therefore$ Current in upper arm $= 2.8 \times \frac{21}{21+7} = 2.1\,\text{A}$

26. Answer (2)

$|\varepsilon| = L \times \frac{di}{dt}$

$5 = L \times \frac{(3-2)}{10^{-3}} \quad \therefore L = 5 \times 10^{-3}\,\text{H} = 5\,\text{mH}$

27. Answer (3)

Maximum energy dissipation occurs at resonance

$f = \frac{1}{2\pi}\sqrt{\frac{1}{LC}} = \frac{1}{2\times22} \times \frac{7}{\sqrt{25\times10^{-3}\times4\times10^{-4}}}$

$= \frac{7\times10^3}{44\sqrt{10}} = 50.3\,\text{Hz}$

28. Answer (1)

Times from max to mean is $\frac{T}{4}$

$t = 0.17 = \frac{T}{4}$

$T = 0.68$ s

$f = \frac{1}{T} = \frac{1}{0.68} = 1.47$ Hz

29. Answer (4)

Permissible error $= \frac{\Delta t}{t} \times 100 = \frac{0.5}{25} \times 100 = 2\%$

30. Answer (2)

Work function $\phi_0 = h\nu_0$

$\frac{1}{2}mv^2 = h\nu - h\nu_0$ ...(i)

$\therefore \quad \frac{1}{2}mv^2 = h(2\nu_0) - h\nu_0 = h\nu_0$

$\frac{1}{2}mv'^2 = h(5\nu_0) - h\nu_0 = 4h\nu_0$

Dividing we get

$\left(\frac{v'}{v}\right)^2 = 4 \qquad \therefore v' = 2v$

31. Answer (4)

Shear strain rate $= \frac{v}{x} = \frac{10\ \text{m s}^{-1}}{20} = 0.5\,\text{s}^{-1}$

$\eta = \frac{\text{Shear stress}}{\text{Shear strain}}$

Shear stress = η × Shear strain rate

$= 10^{-3} \times 0.5 = 5 \times 10^{-4}$

32. Answer (2)

$R_{25} = \frac{(220)^2}{25}\Omega\,; I_1 = \frac{25}{220} = 0.113\,\text{A}$

$R_{100} = \frac{(220)^2}{100}\Omega\,; \ I_2 = \frac{100}{220} = 0.454\,\text{A}$

$R_S = R_1 + R_2 = \frac{(220)^2}{20}\Omega \ \therefore I = \frac{300 \times 20}{(220)^2} = 0.124\,\text{A}$

$V_{25} = 0.124 \times 1936 = 240$ V

$\therefore$ 25 W bulb will fuse.

33. Answer (2)

From diagram $\frac{5\lambda}{2} = 20 \quad \therefore \lambda = 8$ cm

Wave velocity $v = n\lambda = 150 \times 8 \times 10^{-2} = 12\ \text{ms}^{-1}$

Average power in string

$P = 2\pi^2 f^2 A^2 \mu v$

$= 2 \times (3.14)^2 \times (150)^2 \times (0.06)^2 \times (0.2 \times 10^{-3}) \times 12$

$P = 3.84$ W

34. Answer (4)

Fleming's left hand rule to create upward force. So direction of magnetic field should be towards east.

35. Answer (3)

$\theta = \frac{1.22\lambda}{d} = \frac{1.22 \times 600 \times 10^{-9}}{2.5 \times 10^{-3}} = 2.93 \times 10^{-4}$ rad.

$x = D \times \theta = 10 \times 10^3 \times 2.93 \times 10^{-4} = 2.93$ m

36. Answer (2)

$\lambda = \frac{h}{mv}$ and $mvr = \frac{nh}{2\pi}$

Equating $\lambda = \frac{2\pi r}{n} = \frac{2\pi \times 0.53}{1}\text{Å}\,; \lambda = 3.33\,\text{Å}$

37. Answer (4)

By Gauss's law : $\oint E.ds = \frac{q}{\varepsilon_0}$

$E \times 4\pi r^2 = \frac{1}{\varepsilon_0}\int_0^r kr^a.4\pi r^2 dr$

$E = \frac{k}{\varepsilon_0 r^2}\left[\frac{r^{a+3}}{a+3}\right] \ \therefore E = \frac{k(r^{a+1})}{a+3}$

$E_{r=R/2} = \frac{1}{8}(E)_R$

$\frac{k}{\varepsilon_0(a+3)}\left[\frac{R}{2}\right]^{a+1} = \frac{1}{8}\frac{k}{\varepsilon_0(a+3)}(R)^{a+1}$

$\left(\frac{R}{2}\right)^{a+1} = \frac{1}{8} \times (R)^{a+1}$

$\Rightarrow 2^{a+1} = 8\,,\ a = 2$

38. Answer (1)

$x = \frac{(2n+1)\lambda D}{2a}$

For red light $x = \frac{(4+1)D}{2a} \times 6500\,\text{Å}$

For other light $x = \frac{(6+1)D}{2a} \times \lambda\,\text{Å}$

Equating we get

$5 \times (6500) = 7 \times \lambda$

$\lambda = \frac{5}{7} \times 6500 = 4643\,\text{Å}$

39. Answer (3)

$$\text{Energy} = \frac{12400}{6670} \times 1.6 \times 10^{-19}\ \text{J}$$

$$\text{No. of photon} = \frac{P}{E}$$

$$= \frac{9 \times 10^{-3} \times 6670}{12400 \times 1.6 \times 10^{-19}} = 3 \times 10^{16}$$

40. Answer (4)

Resistance of intrinsic semiconductors falls with rise in temperature.

41. Answer (4)

$$\sin c = \frac{1}{\mu}$$

$$\sin i > \frac{1}{\mu} \quad \therefore \mu > \frac{1}{\sin i}$$

$$\mu > \frac{1}{\sin 45^\circ} = \sqrt{2} \quad \therefore \mu > \sqrt{2}$$

Only option (4) is correct for TIR

42. Answer (4)

Since magnetic field is uniform, no change in magnetic flux linked with cylindrical wire and no current will be induced.

43. Answer (3)

$$\text{Original apparent depth} = \frac{12.5}{1.33} = 9.4\ \text{cm}$$

When other liquid used

$$\text{apparent depth} = \frac{12.5}{1.63} = 7.67\ \text{cm}$$

So shift to focus microscope again

$= 9.4 - 7.67 = 1.73$ cm

44. Answer (1)

For real image

$$m = \frac{v}{u} = -4 \qquad \therefore u = \frac{-v}{4}$$

$|v| + |u| = 150$

$$\frac{v}{4} + v = 150 \qquad \therefore v = 120$$

$$u = \frac{-120}{4} = -30\,\text{cm}$$

$$\therefore f = \frac{u \times v}{u - v} = \frac{-30 \times 120}{-30 - 120} = 24\ \text{cm}.$$

45. Answer (1)

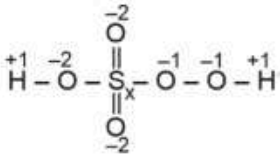
$$L = f_0 + \frac{f_e D}{f_e + D} = 50 + \frac{5 \times 25}{5 + 25}$$

$$= 50 + \frac{25}{6} = 50 + 4.16$$

$\Rightarrow L = 54.16$ cm

CHEMISTRY

46. Answer (3)



Let the oxidation number of sulphur be x.

$\therefore$ $x - 6 = 0$

$x = +6$

47. Answer (3)

Mixture of weak acid and strong base can form an acidic buffer when amount of weak acid is greater than strong base.

48. Answer (3)

$$2NO(g) \rightleftharpoons N_2(g) + O_2(g)$$

$\Delta n_{(g)} = 2 - 2 = 0$

$\therefore$ $K_P = K_C(RT)^{\Delta n_{(g)}}$

$K_P = K_C$

49. Answer (1)

$$\text{At eq.} \quad Ag_2CrO_4(s) \rightleftharpoons \underset{2s}{2Ag^+} + \underset{s}{CrO_4^{2-}}$$

$K_{sp} = (2s)^2 \times s \quad \left(\because \ K_{sp} = [Ag^+]^2 \left[CrO_4^{2-}\right]\right)$

$= 4s^3 = 1.1 \times 10^{-12}$

$s = 6.5 \times 10^{-5}$ mol/litre

50. Answer (2)

$$\text{Total kinetic energy} = \frac{3}{2} nRT$$

$$= \frac{3}{2} \times \frac{40}{20} \times RT = 3RT$$

51. Answer (3)

Norenthindrone : a synthetic progesterone derivative, has antifertility action.

52. Answer (2)

Isoprene (2-methyl-1, 3-butadiene) is a monomer of natural rubber.

53. Answer (2)

Poly β-hydroxybutyrate – co-β-hydroxy valerate (PHBV) is a biodegradable polymer.

54. Answer (4)

Valine is an essential amino acid.

55. Answer (2)

Scurvy causes due to deficiency of vitamin C.

56. Answer (4)

$$n_{H_2O} = \frac{36}{18} = 2 \text{ moles}$$

$$n_{CH_4} = \frac{64}{16} = 4 \text{ moles}$$

$$n_{CO_2} = \frac{44}{44} = 1 \text{ mole}$$

$$n_{H_2} = \frac{10}{2} = 5 \text{ moles}$$

57. Answer (3)

$$E_2 = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J/atom}$$

$$E_2 = -2.18 \times 10^{-18} \times \frac{1}{4}$$

$E_2 = -0.545 \times 10^{-18}$ J/atom

58. Answer (2)

$PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$

$\Delta n_{(g)} = 1$

$\therefore \quad \Delta H = \Delta U + \Delta n_{(g)}RT$

$\Delta H = \Delta U + RT$

Hence $\Delta H > \Delta U$

59. Answer (1)

$$NH_4NO_3 \xrightarrow{\text{Heat}} N_2O + 2H_2O$$

60. Answer (2)

CCl_4 is inert towards hydrolysis under ordinary conditions because of non-availability of vacant *d*-orbitals on carbon atom.

61. Answer (4)

$$\underset{\text{(Cold and dilute)}}{2NaOH} + Cl_2 \rightarrow NaOCl + NaCl + H_2O$$

62. Answer (1)

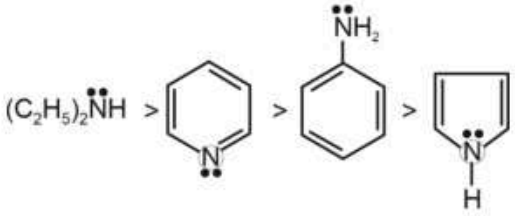
The ion which have vacant '*d*' orbitals or fully filled '*d*' orbitals are generally colourless.

63. Answer (1)

$\Rightarrow$ When electron density increases on nitrogen atom, basic strength increases.

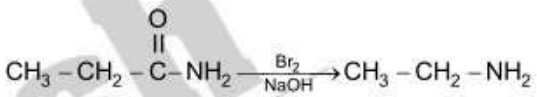
$\Rightarrow$ Due to delocalisation of electron of nitrogen, its basic strength decreases.

$\therefore$ Correct order of basic strength in water is

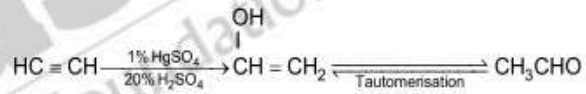


64. Answer (2)

Hoffmann bromamide degradation reaction



65. Answer (2)



66. Answer (3)

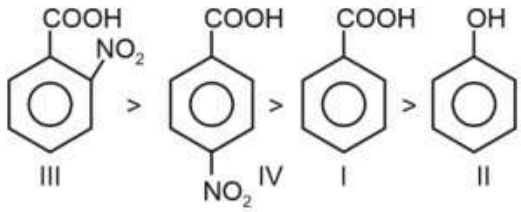
Hell-Volhard-Zelinsky reaction



67. Answer (1)

Electron withdrawing groups increases the acidic strength.

$\therefore$ Correct order of acidic strength is



68. Answer (2)

EAN = 27 – 0 + 8 = 35

69. Answer (3)

Hyperconjugation and resonance increases the stability of free-radical.

∴ Correct order of stability of free radicals is

$CH_2=CH-\dot{C}H_2 > CH_3-\overset{\bullet}{\underset{CH_3}{\underset{|}{C}}}-CH_3 > CH_3-\overset{\bullet}{\underset{CH_3}{\underset{|}{C}}}H > CH_3-\dot{C}H_2$

70. Answer (3)

$CH_3-CH_2-\overset{2}{\underset{\underset{CH_2}{\underset{1}{\|}}}{C}}-\overset{3}{CH_2}-\overset{4}{CH_2}-\overset{5}{CH_3}$

2-ethylpent-1-ene

71. Answer (3)

$[Cr(NH_3)_3Cl_3]$ shows G.I. and have fac and mer isomer.

72. Answer (2)

For 11200 mL of O_2 at STP, 2F is required

∴ 63.5 g of copper will be deposited.

73. Answer (1)

$t_{1/2} \propto [R]_0^{1-n}$; n = order of reaction

when n = 1; first order

∴ $t_{1/2} \propto [R]_0^0$; half life of first order is independent of $[R]_0$.

74. Answer (1)

Cloud is a colloidal solution of liquid in gas.

75. Answer (3)

For B.C.C.

Packing fraction = $\frac{2\times(4/3)\pi r^3}{(4r/\sqrt{3})^3} = \frac{\sqrt{3}\,\pi}{8}$

76. Answer (4)

Molality is independent of temperature because it does not has any volume term in its expression.

77. Answer (3)

For isotonic solution, $\pi_1 = \pi_2$

$\frac{n_1RT}{V_1} = \frac{n_2RT}{V_2}$

$\Rightarrow \frac{18}{180\times1} = \frac{5\times1000}{M_2\times100}$;

$\Rightarrow M_2 = 500\ g\ mol^{-1}$

78. Answer (2)

Isostructural species have the same shape so I_3^- and XeF_2 are isostructural of each other.

79. Answer (2)

If number of electrons increases in antibonding stability of species decreases.

$N_2^-(15): \sigma1s^2\sigma^*1s^2\sigma2s^2\sigma^*2s^2\pi2p_x^2 = \pi2p_y^2\sigma2p_z^2\pi^*2p_x^1$

∴ $B.O. = \frac{10-5}{2} = 2.5$

$N_2^+(13): \sigma1s^2\sigma^*1s^2\sigma2s^2\sigma^*2s^2\pi2p_x^2 = \pi2p_y^2\sigma2p_z^1$

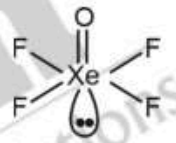
∴ $B.O. = \frac{9-4}{2} = 2.5$

$N_2(14): \sigma1s^2\sigma^*1s^2\sigma2s^2\sigma^*2s^2\pi2p_x^2 = \pi2p_y^2\sigma2p_z^2$

∴ $B.O. = \frac{10-4}{2} = 3$

Correct order of stability is $N_2 > N_2^+ > N_2^-$

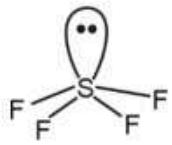
80. Answer (3)



Hybridisation : sp^3d^2

Shape: Square pyramidal

81. Answer (1)



It contains 4σ bonds and one lone pair of electrons.

82. Answer (2)

O = N⁺ = O

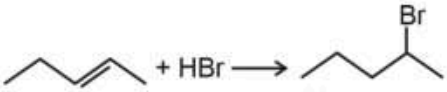
∴ Bond angle = 180°.

83. Answer (4)

∴ Strong electron releasing groups make benzene more reactive towards electrophilic substitution reaction.

84. Answer (2)

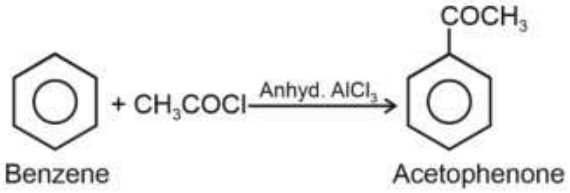
It is electrophilic addition via stable carbocation.



Major product

85. Answer (3)

It is F.C. Acylation.



86. Answer (4)

Common component of oxidising smog are Nitric oxide, O_3 and PAN because they are good oxidising agents.

87. Answer (3)

A mixture of CO and N_2 is called producer gas.

88. Answer (2)

Except Li other alkali metal nitrates decompose to give corresponding nitrite.e.g.

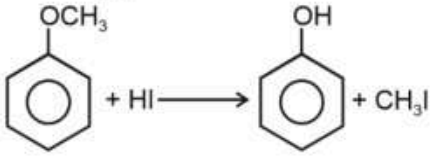
$2NaNO_3 \rightarrow 2NaNO_2 + O_2$

$4LiNO_3 \rightarrow 2Li_2O + 4NO_2 + O_2$

89. Answer (4)

Compound with most stable carbocation intermediate is most reactive towards S_N1 reaction.

90. Answer (2)



(Phenol)

O – CH_3 bond is weaker than O – C_6H_5 (with double bond character)

Hence I^- attacks (O – CH_3) to form CH_3I and phenol.

BOTANY

91. Answer (2)

Ammonia is first oxidised by the bacteria *Nitrosomonas* and/or *Nitrococcus*.

92. Answer (2)

In secondary or biological treatment of sewage, microbes consume the major part of the organic matter in the effluent.

93. Answer (2)

Chlorella, *Spirogyra*, wheat and rice plants are C_3-plants. Maize, sorghum and sugarcane are C_4-plants. RuBP – 5-carbon ketose sugar.

94. Answer (1)

RNA of the viroids is of low molecular weight.

95. Answer (1)

Paul Ehrlich analogy work is popularly known as the Rivet popper hypothesis.

96. Answer (4)

Majority of entomophillous flowers are large, colourful and rich in nectar. To sustain animal visits, the flowers provide rewards to the animals.

97. Answer (3)

Stamens are epipetalous.

98. Answer (1)

In heterosporous pteridophytes, male gametophyte as well as female gametophyte are separate and independent.

99. Answer (3)

Wildlife sanctuary – *In situ* conservation

Pollen bank – *Ex situ* conservation

100. Answer (4)

Archaebacteria differ from eubacteria in having a different cell wall structure; this feature help them to survive in extreme conditions.

101. Answer (1)

Leeuwenhoek examined a number of single celled organisms e.g. bacteria, protozoa, RBC, sperms etc.

102. Answer (4)

AB		B		
$I^A I^B$	X	$I^B I^o$		Phenotype = AB, A, B = 3
$^A I\ I^B$	$^A I\ I^o$	$^B I\ I^B$	$^B I\ I^o$	Genotype = 4

103. Answer (1)

Dictyota is a brown algae with fucoxanthin.

104. Answer (2)

In diplontic life cycle, sporophyte is dominant and independent whereas gametophytic phase is of few cells. Here gametic meiosis occurs.

105. Answer (3)

Collenchyma cells are thickened at the corner due to deposition of cellulose, pectin and hemicellulose.

106. Answer (4)

In monocot stem, the ground tissues is not differentiated into endodermis, pericycle and pith.

107. Answer (2)

In honey bee, haplo-diploid sex determination system is operated. Male is haploid and female is diploid with 16 and 32 chromosomes respectively.

108. Answer (3)

Characteristic palm crease is feature of individual affected with Down's syndrome.

109. Answer (2)

Human skin colour is an example of polygenic inheritance, where environment influences the expression of traits.

110. Answer (2)

Family, is a group of related genera with still less number of similarities as compared to genus and species. Solanaceae is a family.

111. Answer (3)

Rosette of leaves and tuft of roots at each node is present in aquatic plants like *Pistia* and *Eichhornia*.

112. Answer (3)

Placentation is axile but when septa are absent, the placentation is free central.

113. Answer (4)

Seed coat is membranous, endosperm is bulky and embryo is small, situated in a groove at one end of the endosperm.

114. Answer (1)

Both centrioles of a centrosome lie perpendicular to each other in which each has an organisation like cartwheel.

115. Answer (4)

Somatic cell = 2C, after S-phase DNA = 4C

Meiotic product = Gamete = 1C = 12 picogram

At metaphase 4C = 12 x 4 = 48 picogram

116. Answer (1)

Chloroplast length = 5 – 10 μm

Mitochondria = Double membranous

Diameter of cisternae of Golgi complex

= 0.5 – 1.0 μm

117. Answer (4)

Chromosomes are arranged at the metaphasic plate during metaphase. Centromeres are directed towards the poles at anaphase.

118. Answer (3)

Movement of water through semi-permeable membrane takes place from higher ψ_w to lower ψ_w.

119. Answer (2)

Inner wall of each guard cell, towards the stomatal pore is thick and elastic.

120. Answer (1)

There are 17 essential elements.

121. Answer (4)

Cyclic photophosphorylation is observed in light of wavelengths beyond 680 nm.

122. Answer (3)

1 Acetyl Co. A through TCA and ETS produces 12 ATP.

4 Acetyl Co. A = 12 × 4 = 48 ATP

123. Answer (1)

Only angiosperms show double fertilization.

124. Answer (2)

Natural cytokinins are synthesised in regions where rapid cell division occurs for example, root apices, shoot apices, young fruit etc.

125. Answer (3)

Electron microscope

126. Answer (4)

DNA finger printing works on the principle of polymorphism in DNA sequences, to find out variations in individuals of a population.

127. Answer (2)

The vectors provide the origin of replication.

128. Answer (1)

Most of the amino acids are coded by more than one codon, hence the code is degenerate.

129. Answer (3)

Pusa Gaurav is variety of crop *Brassica*. (Rapeseed mustard) resistant to aphids.

130. Answer (3)

Species composition and stratification are main structural components of ecosystem.

131. Answer (2)

High concentrations of DDT disturbs calcium metabolism in birds, which causes thinning of egg shell and their premature breaking.

132. Answer (3)

Catalytic converter, converts unburnt hydrocarbons into carbon dioxide and carbon monoxide which is converted to carbon dioxide.

133. Answer (4)

Birth rate, death rates, sex ratio etc are population attributes.

134. Answer (2)

Chemosynthetic autotrophic bacteria oxidise various inorganic substances and play important role in recycling of certain nutrients.

135. Answer (3)

Banana is one of the parthenocarpic fruit. It can be induced through application of PGR.

ZOOLOGY

136. Answer (2)

Spinal cord passes through the neural canal of vertebral column.

137. Answer (2)

All ribs articulate dorsally to thoracic vertebrae of vertebral column.

138. Answer (2)

Biceps of arms are example of skeletal muscles.

139. Answer (3)

Salpa belongs to subphylum Urochordata.

140. Answer (4)

A pair of anal cerci is found in both male and female cockroach.

141. Answer (2)

In *Periplaneta*, the thorax region possess three pairs of legs and two pairs of wings.

142. Answer (3)

External ear openings are absent in reptiles.

143. Answer (1)

Presence of air/swim bladder is a characteristic of osteichthyes.

144. Answer (2)

Midbrain possesses 4 lobes in their dorsal side called corpora quadrigemina.

145. Answer (1)

Hypothalamus lies at the base of thalamus.

146 Answer (3)

Vitreous chamber is present between lens and the retina.

147. Answer (2)

Tympanum or ear drum is present at the end of external auditory canal.

148. Answer (2)

TCT and PTH collectively regulate the blood Ca^{+2} level.

149. Answer (2)

Graves' disease is hypothyroidism in adult stage, characterised by protrusion of the eye balls.

150. Answer (2)

Glucagon and insulin are peptide hormones.

151. Answer (2)

Terrestrial amphibians, mammals and mostly marine fishes and ureotelic.

152. Answer (4)

Each kidney of an adult human measures 10-12 cm in length and 5-7 cm in width.

153. Answer (2)

Tubular secretion is a important step of urine formation. It is helpful in maintenance of ionic and acid-base balance of body and occurs in all parts of nephron except loop of Henle and Bowmann's capsule.

154. Answer (3)

Person with O blood group can donate the blood to any persons with B, AB, A and O blood group.

155. Answer (1)

Cardiac output = Heart rate × Stroke volume

5040 = 72 × Stroke volume

$$\text{Stroke volume} = \frac{5040 \text{ ml / minute}}{72 \text{ / minute}} = 70 \text{ ml}$$

156. Answer (2)

Intra-pulmonary pressure slightly exceeds the atmospheric pressure during expiration.

157. Answer (1)

FRC = ERV + RV

158. Answer (2)

DNase acts on DNA whereas RNase acts on RNA in small intestine.

159. Answer (3)

Oxyntic cells secrete Castle's intrinsic factor which helps in absorption of vitamin B_{12} from jejunum.

160. Answer (2)

IgA has four paratopes.

161. Answer (3)

Cocaine interferes with the transport of the neuro-transmitter dopamine.

162. Answer (3)

Infected female *Anopheles* mosquito is vector for malaria in humans.

163. Answer (4)

Out-crossing is useful in overcoming inbreeding depression.

164. Answer (1)

White revolution is associated with enhancement of milk production.

165. Answer (2)

The panlindromic base sequence for *Bam* HI is

5′ … GGATCC …3′

3′ … CCTAGG …5′

166. Answer (2)

Optimum temperature for annealing step is 56°C (50°C to 60°C)

167. Answer (3)

Insertional inactivation in *lac* z gene make it non functional, now it can't form enzyme beta galactosidase.

168. Answer (2)

If we intake proteins orally they will get digested in stomach and small intestine.

169. Answer (4)

Vector in *Agrobacterium* is useful of dicot plants.

170. Answer (3)

Darwin finches and Australian marsupials are examples of adaptive radiations.

171. Answer (3)

$p = 0.06$

So $q = 0.4$

Therefore the frequency of heterozygous is $2pq = 0.48$

172. Answer (1)

Analogous organs are a result of convergent evolution.

173. Answer (3)

Implants have mechanism of action similar to oral pills.

174. Answer (2)

In AI (Artificial insemination), sperms are transferred into the female reproductive tract.

ZIFT : Zygote intra fallopian transfer

175. Answer (2)

Chromosome number in meiocyte of dog is 78.

176. Answer (2)

Cockroach is a dioecious organism.

177. Answer (2)

The process of transformation of spermatid into sperm is called spermiogenesis.

178. Answer (3)

Polar bodies, secondary spermatocytes and secondary oocytes are haploid structures.

179. Answer (3)

Follicular cells of thyroid follicles are made up of cuboidal epithelium.

180. Answer (1)

Glycogen is a branched homopolysaccharides of glucose.

❑ ❑ ❑

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